IMAGE OF A HOMOLOGICAL SPECTRUM MAP AS SUPPORT OF THE ADJOINT UNIT

PAUL BALMER

Editorial note. Source726 writes the Yoneda commutation with a hat on the final F/U although these final functors act on the triangulated categories. The explicit diagram735–741 and the typed Kan formulas728–732 identify the intended unhatted functors. The author text is retained verbatim; this note explains the local notation only.

Editorial note. Source705 writes the internal-Hom kernel using E_B, whereas559 and637 define the same kernel in the module category using the Yoneda image hat(E_B). The source is retained; the omitted hat is read according to those displayed definitions.

Allowed standard prerequisites (editorial).

- Restricted Yoneda, Day convolution and pure-injective representation of injectives for a rigidly-compactly generated category. Foundational module-category facts, including exact internal Hom into injectives, are provided. The local weak-ring/maximal-ideal and residue-pullback arguments are proved, not assumed.
- Gabriel localization of Grothendieck categories and finitely presented generation. For a Serre tensor ideal B of finitely presented T^c-modules and its pure-injective weak ring E_B, B={M finitely presented: hat(E_B) tensor M=0} and Loc(B)=Ker(hat(E_B) tensor -); the quotient is locally coherent and a subobject of a finitely presented object is the filtered colimit of its finitely presented subobjects. These are the pre-existing BKS2019 results recalled423–443, especially433; they are permitted input, not a new claim of this article.. This pre-existing tensor-kernel recovery and finite-presentation infrastructure is supplied. The article-specific comparison E_B=E_C, uniqueness of the maximal localizing ideal, maximality of residue pullbacks, the summand construction and both image inclusions remain required proof obligations.
- Brown–Neeman representability, adjunctions and projection formula for coproduct-preserving tt functors. General adjoint existence and projection formula; lifting residue categories and detecting the image are not supplied.
- Homological nilpotence theorem for a map from a compact object. Used only in the additional weak-ring detection conclusion705 retained with its author proof; image equality uses the preceding support equality, not a new proof of the nilpotence theorem.

0.1. **Hypothesis.** Let $\mathcal{T}$ be a *'big' tensor-triangulated category*, meaning a rigidly-compactly generated one, as in [7]. So $\mathcal{T}$ admits small coproducts; its subcategory $\mathcal{T}^c$ of compact objects coincides with that of rigid (strongly-dualizable) objects; $\mathcal{T}^c$ is essentially small and generates $\mathcal{T}$ as a localizing subcategory.

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0.2. **Definition.** We say that an object A in $\mathcal{T}$ with a map $\eta\colon \mathbb{1} \to A$ (its 'unit') is a *weak ring* if $A \otimes \eta\colon A \to A \otimes A$ is a split monomorphism (whose retraction $A \otimes A \to A$ can be thought of as a unital non-associative multiplication on A).

0.3. **Recollection.** The *homological spectrum* $\operatorname{Spc}^{\mathrm{h}}(\mathcal{T}^c)$ consists of all maximal Serre $\otimes$-ideals $\mathcal{B}$ of the abelian subcategory $\mathcal{A}^{\mathrm{fp}} = \text{mod-}\mathcal{T}^c$ of finitely presented $\mathcal{T}^c$-modules. We call those $\mathcal{B}$ the *homological primes* of $\mathcal{T}$. Note that they only depend on the subcategory $\mathcal{T}^c$. For each $\mathcal{B} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{T}^c)$, its preimage in $\mathcal{T}^c$ under Yoneda

$$\phi(\mathcal{B}) = \mathbb{h}^{-1}(\mathcal{B}) \cap \mathcal{T}^c = \big\{ x \in \mathcal{T}^c \,\big|\, \hat{x} \in \mathcal{B} \big\}$$

is a (triangular) prime ideal in $\mathcal{T}^c$. This defines a surjection $\phi\colon \operatorname{Spc}^{\mathrm{h}}(\mathcal{T}^c) \twoheadrightarrow \operatorname{Spc}(\mathcal{T}^c)$ by [4, Cor. 3.9]. Each $\mathcal{B} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{T}^c)$ yields a coproduct-preserving homological tensor-functor $\bar{\mathbb{h}}_{\mathcal{B}}\colon \mathcal{T} \to \bar{\mathcal{A}}_{\mathcal{B}} = \mathcal{A}/\operatorname{Loc}(\mathcal{B})$ as in (1.13).

0.4. **Definition.** The *(homological) support* of an arbitrary object X in the big tt-category $\mathcal{T}$ is the following subset of the homological spectrum $\operatorname{Spc}^{\mathrm{h}}(\mathcal{T}^c)$:

$$\operatorname{Supp}(X) := \big\{ \mathcal{B} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{T}^c) \,\big|\, [X, E_{\mathcal{B}}]_{\mathcal{T}} \neq 0 \big\}.$$

1. Yoneda and modules

Many readers can safely skip this section and refer back to it as needed, especially those familiar with the module category

$$\mathcal{A} := \text{Mod-}\mathcal{T}^c = \operatorname{Add}((\mathcal{T}^c)^{\mathrm{op}}, \mathrm{Ab})$$

of additive contravariant functors from $\mathcal{T}^c$ to abelian groups.

1.1. **Recollection.** The abelian category $\mathcal{A} = \text{Mod-}\mathcal{T}^c$ is a Grothendieck category, whose subcategory of finitely presented objects $\mathcal{A}^{\mathrm{fp}} = \text{mod-}\mathcal{T}^c$ coincides with the Freyd envelope of $\mathcal{T}^c$ ([23, Chap. 5]). The (restricted) Yoneda embedding $\mathbb{h}$

$$\begin{array}{ccc} \mathcal{T}^c & \xhookrightarrow{\ \mathbb{h}\ } & \text{mod-}\mathcal{T}^c = \mathcal{A}^{\mathrm{fp}} \\ \big\downarrow & & \big\downarrow \\ \mathcal{T} & \xrightarrow{\ \mathbb{h}\ } & \text{Mod-}\mathcal{T}^c = \mathcal{A} \end{array}$$

is defined by $\mathbb{h}(X) = \hat{X}$ where $\hat{X} = \operatorname{Hom}_{\mathcal{T}}(-, X)_{|\mathcal{T}^c}$. This functor $\mathbb{h}\colon \mathcal{T} \to \mathcal{A}$ is homological (maps distinguished triangles to exact sequences), preserves coproducts and is universal among those ([19, Cor. 2.4]). It is also conservative.

Restricted-Yoneda $\mathbb{h}$ is fully faithful on $\mathcal{T}^c$, and identifies the latter with finitely presented projective objects in $\mathcal{A}$. Every (big) object $M \in \mathcal{A}$ is a filtered colimit of finitely presented objects. (Indeed $\mathcal{A}$ is *locally coherent*; see [15, A.7].) Also, every object $M \in \mathcal{A}$ is a quotient of a coproduct $\coprod_{i\in I} \hat{x}_i$ of rigid-compact objects $x_i \in \mathcal{T}^c$. For an object $P = \coprod_{i\in I} x_i$ with all $x_i \in \mathcal{T}^c$ (or a summand of such a coproduct) and for $Y \in \mathcal{T}$ arbitrary, restricted-Yoneda yields an isomorphism

$$\mathbb{h}\colon \operatorname{Hom}_{\mathcal{T}}(P, Y) \xrightarrow{\sim} \operatorname{Hom}_{\mathcal{A}}(\hat{P}, \hat{Y}). \tag{1.2}$$

Hence all projectives in $\mathcal{A}$ are $\hat{P}$ for P a summand of some $\coprod_{i\in I} x_i$ with all $x_i \in \mathcal{T}^c$.

The category $\mathcal{A}$ also has enough injectives and they also come from $\mathcal{T}$. By [19], they are all of the form $\hat{E}$ for a unique $E \in \mathcal{T}$, called a *pure-injective*. For every object X and every pure-injective E in $\mathcal{T}$, restricted-Yoneda gives an isomorphism

$$\mathbb{h}\colon \operatorname{Hom}_{\mathcal{T}}(X, E) \xrightarrow{\sim} \operatorname{Hom}_{\mathcal{A}}(\hat{X}, \hat{E}). \tag{1.3}$$

1.4. **Recollection.** An essential feature of the module category is its tensor product, obtained by Day convolution, and discussed in [15, App. A]. This tensor is colimit-preserving in each variable, in particular it is right-exact. It makes the restricted-Yoneda functor (not just the part on $\mathcal{T}^c$) into a tensor functor $\mathbb{h}\colon \mathcal{T} \to \mathcal{A}$ and every object in the image of $\mathcal{T}$ is $\otimes$-flat, i.e., $\hat{X} \otimes -$ is exact for all $X \in \mathcal{T}$.

As a consequence of this, all projective objects of $\mathcal{A}$ and, perhaps more remarkably, all injective objects of $\mathcal{A}$ are $\otimes$-flat.

By general Grothendieck-category theory $\mathcal{A}$ is then *closed* monoidal, i.e., it admits an internal-hom functor $[-,-]_{\mathcal{A}}\colon \mathcal{A}^{\mathrm{op}} \times \mathcal{A} \longrightarrow \mathcal{A}$. Beware that $\mathbb{h}$ might not be a closed functor but one can easily upgrade (1.3) into an isomorphism

$$\mathbb{h}([X,E]_{\mathcal{T}}) \cong [\hat{X},\hat{E}]_{\mathcal{A}} \tag{1.5}$$

for every $X, E \in \mathcal{T}$ with E pure-injective, by testing via $\mathrm{Hom}_{\mathcal{A}}(\hat{c},-)$ with $c \in \mathcal{T}^c$ and using that $\mathbb{h}$ is a tensor functor.

1.6. **Recollection.** Let $\mathcal{C}$ in $\mathcal{A}$ be a localizing subcategory (i.e., closed under coproducts, extensions, subobjects and quotients). We have a Gabriel quotient $\mathcal{A}/\mathcal{C}$

$$\begin{array}{c} \mathcal{A} \\ Q_{\mathcal{C}} \big\downarrow \big\uparrow R_{\mathcal{C}} \\ \mathcal{A}/\mathcal{C} \end{array} \tag{1.7}$$

where $Q_{\mathcal{C}}$ is the universal exact functor with kernel $\mathcal{C}$, its right adjoint $R_{\mathcal{C}}$ is fully-faithful and $Q_{\mathcal{C}} \circ R_{\mathcal{C}} \cong \mathrm{Id}_{\mathcal{A}/\mathcal{C}}$. We shall only consider subcategories $\mathcal{C}$ that are $\otimes$-ideal ($\mathcal{A} \otimes \mathcal{C} \subseteq \mathcal{C}$), in which case $\mathcal{A}/\mathcal{C}$ inherits a unique tensor structure such that $Q_{\mathcal{C}}$ is a tensor functor. When $\mathcal{C}$ is clear from the context, we often write $\bar{X}$ instead of $Q_{\mathcal{C}}(\hat{X})$ for $X \in \mathcal{T}$. We also write $\bar{\mathcal{A}}_{\mathcal{C}}$ (or just $\bar{\mathcal{A}}$) for $\mathcal{A}/\mathcal{C}$ and

$$\bar{\mathbb{h}}_{\mathcal{C}}\colon \mathcal{T} \longrightarrow \bar{\mathcal{A}}_{\mathcal{C}} = \mathcal{A}/\mathcal{C} \tag{1.8}$$

for the composite $Q_{\mathcal{C}} \circ \mathbb{h}$, a coproduct-preserving cohomological tensor-functor.

Those quotients $\mathcal{A}/\mathcal{C}$ inherit a number of properties that hold for $\mathcal{A}$.

1.9. **Proposition.** *With notation as in Equation* (1.6)*, we have:*

(a) *The Gabriel quotient $\bar{\mathcal{A}} = \mathcal{A}/\mathcal{C}$ is a Grothendieck category, that is closed monoidal. Its tensor product preserves colimits in each variable.*
(b) *Every $X \in \mathcal{T}$ has $\otimes$-flat image $\bar{X} = Q_{\mathcal{C}}(\hat{X})$ in $\bar{\mathcal{A}}$. If $x \in \mathcal{T}^c$ then $\bar{x}$ is rigid.*
(c) *Every object of $\bar{\mathcal{A}}$ is a quotient of a coproduct $\bigsqcup_{i\in I} \bar{x}_i$ with all $x_i \in \mathcal{T}^c$.*
(d) *Every injective object of $\bar{\mathcal{A}}$ is of the form $\bar{E}$ for some pure-injective E in $\mathcal{T}$, uniquely characterized by the property $\hat{E} \cong R_{\mathcal{C}}(\bar{E})$ in $\mathcal{A}$. In particular, $\bar{E}$ is $\otimes$-flat. Furthermore, $\bar{\mathbb{h}}_{\mathcal{C}}\colon \mathcal{T} \to \bar{\mathcal{A}}$ induces, for every $X \in \mathcal{T}$, an isomorphism*
$$\mathrm{Hom}_{\mathcal{T}}(X,E) \xrightarrow{\sim} \mathrm{Hom}_{\bar{\mathcal{A}}}(\bar{X},\bar{E}).$$
(e) *The right adjoint $R_{\mathcal{C}}\colon \bar{\mathcal{A}} \to \mathcal{A}$ is lax-monoidal and closed; more precisely for every $M \in \mathcal{A}$ and $N \in \bar{\mathcal{A}}$, if we denote by $[-,-]$ the internal-homs, we have*
$$[M, R_{\mathcal{C}}(N)]_{\mathcal{A}} \cong R_{\mathcal{C}}([Q_{\mathcal{C}}(M), N]_{\bar{\mathcal{A}}}).$$
(f) *For every injective $\bar{E}$ in $\bar{\mathcal{A}}$, the internal-hom $[-,\bar{E}]_{\bar{\mathcal{A}}}\colon \bar{\mathcal{A}}^{\mathrm{op}} \to \bar{\mathcal{A}}$ is exact.*

Proof. These claims follow from [15, App. A] and [14, §2 & App. A]. Let us prove (f), for [14, Lem. 2.8] is only stated for $\mathcal{A}$. We have

$$[-,\bar{E}]_{\bar{\mathcal{A}}} \circ Q_{\mathcal{C}} = [Q_{\mathcal{C}}(-),\bar{E}]_{\bar{\mathcal{A}}} \cong Q_{\mathcal{C}} R_{\mathcal{C}}([Q_{\mathcal{C}}(-),\bar{E}]_{\bar{\mathcal{A}}}) \underset{\text{(e)}}{\cong} Q_{\mathcal{C}} \circ [-, R_{\mathcal{C}}(\bar{E})]_{\mathcal{A}}$$

and the latter is exact by [14, Lem. 2.8], since $R_{\mathcal{C}}\bar{E}$ is injective in $\mathcal{A}$ itself. As pre-composition with $Q_{\mathcal{C}}$ detects exactness for functors on $\bar{\mathcal{A}}$, we get the result. □

1.10. **Remark.** Note that $\Sigma\colon \mathcal{T} \to \mathcal{T}$ induces an auto-equivalence $\Sigma\colon \mathcal{A} \to \mathcal{A}$ which is isomorphic to $\mathbb{h}(\Sigma\mathbb{1}) \otimes -$. Hence all our $\otimes$-ideals $\mathcal{C}$ in $\mathcal{A}$ are stable under Σ and all quotients inherit a suspension $\Sigma\colon \bar{\mathcal{A}}_{\mathcal{C}} \xrightarrow{\sim} \bar{\mathcal{A}}_{\mathcal{C}}$ such that $Q_{\mathcal{C}}\Sigma \cong \Sigma Q_{\mathcal{C}}$.

An important construction is the one of [14, §3], slightly generalized:

1.11. **Construction.** For $\mathcal{C} \subseteq \mathcal{A}$ localizing $\otimes$-ideal, the injective hull of $\bar{\mathbb{1}}$ in $\bar{\mathcal{A}}_{\mathcal{C}}$ comes from $\bar{\mathbb{h}}_{\mathcal{C}}$, by Equation (1.9) (d). So there exists a morphism

$$\eta_{\mathcal{C}}\colon \mathbb{1} \to E_{\mathcal{C}}$$

in $\mathcal{T}$ such that $\bar{\eta}_{\mathcal{C}} = \bar{\mathbb{h}}_{\mathcal{C}}(\eta_{\mathcal{C}})\colon \bar{\mathbb{1}} \rightarrowtail \bar{E}_{\mathcal{C}}$ is the injective hull. It is characterized by $R_{\mathcal{C}}(\bar{E}_{\mathcal{C}}) \cong \hat{E}_{\mathcal{C}}$ in $\mathcal{A}$. Since $\bar{E}_{\mathcal{C}}$ is flat and injective, $\bar{E}_{\mathcal{C}} \otimes \bar{\eta}_{\mathcal{C}}$ is a split monomorphism in $\bar{\mathcal{A}}_{\mathcal{C}}$, hence by Equation (1.9) (d) again, the retraction exists in $\mathcal{T}$, meaning that $(E_{\mathcal{C}}, \eta_{\mathcal{C}})$ is a *weak ring* in $\mathcal{T}$ in the sense of Equation (0.2).

1.12. **Recollection.** A particular class of localizing $\otimes$-ideals $\mathcal{C} \subseteq \mathcal{A}$ are those generated by Serre $\otimes$-ideals $\mathcal{B} \subseteq \mathcal{A}^{\mathrm{fp}} = \text{mod-}\mathcal{T}^c$ of finitely presented objects (where $\otimes$-ideal only means $\mathcal{A}^{\mathrm{fp}} \otimes \mathcal{B} \subseteq \mathcal{B}$ of course). Explicitly $\mathcal{C} = \mathrm{Loc}(\mathcal{B})$ is the smallest localizing subcategory containing $\mathcal{B}$, which is then automatically $\otimes$-ideal in $\mathcal{A}$. In some notation, typically in indices, we drop the 'Loc' part and write only $\mathcal{B}$, like for instance with the canonical cohomological tensor-functor (1.8):

$$\bar{\mathbb{h}}_{\mathcal{B}}\colon \mathcal{T} \longrightarrow \bar{\mathcal{A}}_{\mathcal{B}} := \mathcal{A}/\operatorname{Loc}(\mathcal{B}). \tag{1.13}$$

Similarly, $E_{\mathcal{B}}$ means $E_{\mathrm{Loc}(\mathcal{B})}$ as in Equation (1.11) for $\mathcal{C} = \mathrm{Loc}(\mathcal{B})$.

Under these additional assumptions, we know more than in Equation (1.9):

1.14. **Proposition.** *With notation as in Equation* (1.12)*, we have:*

(a) *The Grothendieck category $\bar{\mathcal{A}}_{\mathcal{B}}$ is still locally coherent. Its finitely presented objects $\bar{\mathcal{A}}^{\mathrm{fp}} = \mathcal{A}^{\mathrm{fp}}/\mathcal{B}$ coincide with the Gabriel quotient of $\mathcal{A}^{\mathrm{fp}}$ by $\mathcal{B}$.*

(b) *We can recover $\mathcal{B}$ and $\mathrm{Loc}(\mathcal{B})$ from the pure-injective weak ring $E_{\mathcal{B}}$ of Equation* (1.11)*, as $\mathcal{B} = \big\{M \in \mathcal{A}^{\mathrm{fp}} \,\big|\, \hat{E}_{\mathcal{B}} \otimes M = 0\big\}$ and $\mathrm{Loc}(\mathcal{B}) = \mathrm{Ker}(\hat{E}_{\mathcal{B}} \otimes -)$.*

(c) *Every subobject of a finitely presented object in $\bar{\mathcal{A}}_{\mathcal{B}}$ is the colimit of its finitely presented subobjects. In particular, if $I \rightarrowtail \bar{\mathbb{1}}$ in $\bar{\mathcal{A}}_{\mathcal{B}}$ then $I = \operatorname*{colim}_{\text{f.p. } M \rightarrowtail I} M$.*

Proof. All this is in [14, §3]. □

We shall need the following general observation.

1.15. **Lemma.** *Let E be an injective object in $\mathcal{A}$. Then the subcategory $\mathrm{Ker}[-,E]_{\mathcal{A}} = \big\{M \in \mathcal{A} \,\big|\, [M,E]_{\mathcal{A}} = 0\big\}$ is a localizing $\otimes$-ideal of $\mathcal{A}$.*

Proof. This kernel $\mathrm{Ker}[-,E]_{\mathcal{A}}$ is a localizing (in particular, Serre) subcategory because the internal-hom functor $[-,E]_{\mathcal{A}}$ turns coproducts into products and is exact by Equation (1.9) (f) (or [14, Lem. 2.8]). This kernel is closed under tensoring since $[M \otimes N, E]_{\mathcal{A}} \cong [M, [N,E]_{\mathcal{A}}]_{\mathcal{A}}$ vanishes if $[N,E]_{\mathcal{A}}$ does. □

1.16. **Remark.** The same statement holds in any $\bar{\mathcal{A}}_{\mathcal{C}}$, with a similar proof.

1.17. **Remark.** The maximality of $\mathcal{B} \in \mathrm{Spc}^{\mathrm{h}}(\mathcal{T}^c)$ among the Serre $\otimes$-ideals of $\mathcal{A}^{\mathrm{fp}}$ tells us that $\bar{\mathcal{A}}^{\mathrm{fp}}_{\mathcal{B}}$ is 'simple': it has only the two trivial Serre $\otimes$-ideals, zero and $\bar{\mathcal{A}}^{\mathrm{fp}}_{\mathcal{B}}$. (See Equation (1.12) if necessary.) Simplicity has the following consequence.

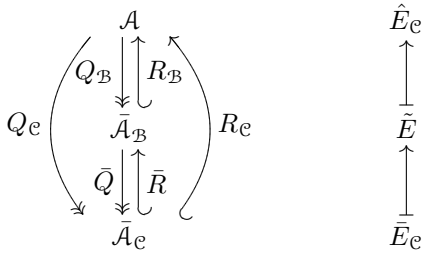
1.18. **Proposition.** *Let $\mathcal{B} \in \mathrm{Spc}^{\mathrm{h}}(\mathcal{T}^c)$ and $A \in \bar{\mathcal{A}}_{\mathcal{B}}$ be a weak ring (Equation* (0.2)*) that is $\otimes$-flat, like $A = \bar{E}$ for some weak ring $E \in \mathcal{T}$. Then either $A = 0$ or its unit is a monomorphism $\bar{\mathbb{1}} \rightarrowtail A$.*

Proof. This is a basic trick in the proof of the Nilpotence Theorem [4]. Consider the exact sequence $\mathrm{Ker}(\eta) \rightarrowtail \bar{\mathbb{1}} \xrightarrow{\eta} A$ in $\bar{\mathcal{A}}_{\mathcal{B}}$. Since A is flat and $A \otimes \eta$ is a (split) monomorphism, we have $A \otimes \mathrm{Ker}(\eta) = 0$. Suppose that $\mathrm{Ker}(\eta) \neq 0$ and let us show that $A = 0$. By Equation (1.14) (c), we know that $\mathrm{Ker}(\eta)$ is the colimit of its finitely presented subobjects. Take a finitely presented $M \neq 0$ with $M \rightarrowtail \mathrm{Ker}(\eta)$. By flatness of A and $A \otimes \mathrm{Ker}(\eta) = 0$, we see that $A \otimes M = 0$. Hence $\mathrm{Ker}(A \otimes -) \cap \bar{\mathcal{A}}^{\mathrm{fp}}_{\mathcal{B}}$ is a non-zero Serre (by flatness) $\otimes$-ideal of $\bar{\mathcal{A}}^{\mathrm{fp}}_{\mathcal{B}}$. As $\bar{\mathcal{A}}^{\mathrm{fp}}_{\mathcal{B}}$ is 'simple' (Equation (1.17)) this forces $\mathrm{Ker}(A \otimes -) \cap \bar{\mathcal{A}}^{\mathrm{fp}}_{\mathcal{B}} = \bar{\mathcal{A}}^{\mathrm{fp}}_{\mathcal{B}}$ to contain $\bar{\mathbb{1}}$, giving $A = 0$. □

1.19. **Remark.** During the year the author spent in Bielefeld working with Krause and Stevenson on [14], we hesitated between maximal Serre $\otimes$-ideals $\mathcal{B} \subset \mathcal{A}^{\mathrm{fp}}$ and maximal localizing $\otimes$-ideals $\mathcal{C} \subset \mathcal{A}$. We opted for the finitely presented ones for the extra properties of Equation (1.14), that turned out to be useful in [14] and later in the proof of the Nilpotence Theorem [4]. It remains an open question to relate the two notions. Let us first see that every localizing $\otimes$-ideal bigger than a homological prime has the 'same' pure-injective weak ring $E_{\mathcal{B}}$.

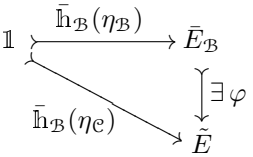
1.20. **Proposition.** *Let $\mathcal{B} \in \mathrm{Spc}^{\mathrm{h}}(\mathcal{T}^{\mathrm{c}})$ and $\mathcal{C} \subset \mathcal{A}$ be a proper localizing $\otimes$-ideal, containing $\mathcal{B}$ or equivalently containing $\mathrm{Loc}(\mathcal{B})$. Let $E_{\mathcal{B}}$ and $E_{\mathcal{C}}$ be the corresponding weak rings of Equation (1.11). Then there is an isomorphism $E_{\mathcal{B}} \simeq E_{\mathcal{C}}$ in $\mathcal{T}$ compatible with the unit maps $\mathbb{1} \to E_{\mathcal{B}}$ and $\mathbb{1} \to E_{\mathcal{C}}$.*

Proof. Since $\mathrm{Loc}(\mathcal{B}) \subseteq \mathcal{C}$, we can perform the Gabriel quotient $\mathcal{A}/\mathcal{C}$ in two steps:



for an intermediate Gabriel localization $\bar{Q} \dashv \bar{R}$. Let $\tilde{E} := \bar{R}(\bar{E}_{\mathcal{C}})$ the injective object in $\bar{\mathcal{A}}_{\mathcal{B}}$ associated to the injective hull $\bar{\mathbb{1}} \rightarrowtail \bar{E}_{\mathcal{C}}$ in $\bar{\mathcal{A}}_{\mathcal{C}}$. By construction of $E_{\mathcal{C}} \in \mathcal{T}$, we have $\hat{E}_{\mathcal{C}} \cong R_{\mathcal{C}}(\bar{E}_{\mathcal{C}}) \cong R_{\mathcal{B}}(\tilde{E})$ and therefore $\tilde{E} \cong Q_{\mathcal{B}}(\hat{E}_{\mathcal{C}}) = \bar{\mathbb{h}}_{\mathcal{B}}(E_{\mathcal{C}})$. Since $E_{\mathcal{C}}$ is a weak ring in $\mathcal{T}$, its image $\tilde{E}$ is a weak ring in $\bar{\mathcal{A}}_{\mathcal{B}}$, that is non-zero since $\bar{Q}(\tilde{E}) \cong \bar{E}_{\mathcal{C}} \neq 0$. By Equation (1.18), the unit $\bar{\mathbb{h}}_{\mathcal{B}}(\eta_{\mathcal{C}})\colon \bar{\mathbb{1}} \rightarrowtail \tilde{E}$ is therefore a monomorphism in $\bar{\mathcal{A}}_{\mathcal{B}}$. On the other hand, $\bar{\mathbb{h}}_{\mathcal{B}}(\eta_{\mathcal{B}})\colon \bar{\mathbb{1}} \rightarrowtail \bar{E}_{\mathcal{B}}$ is by definition the injective hull of $\bar{\mathbb{1}}$ in $\bar{\mathcal{A}}_{\mathcal{B}}$. Hence there exists a commutative diagram in $\bar{\mathcal{A}}_{\mathcal{B}}$

(1.21)



where φ is a split monomorphism. Applying the exact functor $\bar{Q}$ to this diagram and using that $\bar{Q}(\tilde{E}) = \bar{E}_{\mathcal{C}}$ is the injective hull of $\bar{\mathbb{1}}$ in $\bar{\mathcal{A}}_{\mathcal{C}}$, we see that $\bar{Q}(\varphi)$ must be an isomorphism. In summary, $\tilde{E} \cong \bar{E}_{\mathcal{B}} \oplus N$ in $\bar{\mathcal{A}}_{\mathcal{B}}$ for an object N such that $\bar{Q}(N) = 0$. But since $\tilde{E} = \bar{R}(\bar{E}_{\mathcal{C}})$ and $\bar{R}$ is fully faithful, this forces the object N to be in the image of $\bar{R}$ as well (it is '$\bar{R}$-local'). It follows that $N \cong \bar{R}\bar{Q}(N) = 0$. Therefore φ is already an isomorphism $\bar{E}_{\mathcal{B}} \simeq \tilde{E} \simeq \bar{R}(\bar{E}_{\mathcal{C}})$ in $\bar{\mathcal{A}}_{\mathcal{B}}$. By Equation (1.9) (d) (applied to $\bar{\mathcal{A}} = \mathcal{A}/\mathrm{Loc}(\mathcal{B})$) the isomorphism φ in (1.21) comes from $\mathcal{T}$ and is compatible with the units there. □

1.22. **Remark.** Recall from Equation (1.14) (b) that $\mathrm{Loc}(\mathcal{B}) = \mathrm{Ker}(\hat{E}_{\mathcal{B}} \otimes -)$. What Equation (1.20) tells us is that a strictly larger localizing $\otimes$-ideal $\mathrm{Loc}(\mathcal{B}) \subsetneq \mathcal{C} \subsetneq \mathcal{A}$ will share the same pure-injective weak ring $E_{\mathcal{C}} = E_{\mathcal{B}}$ and in particular cannot be equal to $\mathrm{Ker}(\hat{E}_{\mathcal{C}} \otimes -)$. On the other hand, we saw in Lemma 1.15 that there is another way of constructing a localizing $\otimes$-ideal. Let us compare them.

1.23. **Proposition.** *Let $\mathcal{C} \subset \mathcal{A}$ be a localizing $\otimes$-ideal and $E_{\mathcal{C}}$ the associated weak ring (Equation* (1.11)*). Then we have* $\mathrm{Ker}(\hat{E}_{\mathcal{C}} \otimes -) \subseteq \mathcal{C} \subseteq \mathrm{Ker}([-, \hat{E}_{\mathcal{C}}])$.

Proof. The first inclusion can be found in [14]: Every $M \in \mathcal{A}$ admits an injective hull, say $M \rightarrowtail F$, with F necessarily $\otimes$-flat (Equation (1.4)). Suppose that $\hat{E}_{\mathcal{C}} \otimes M = 0$. Down in $\bar{\mathcal{A}}_{\mathcal{C}}$, tensoring $\bar{M} \rightarrowtail \bar{F}$ with $\bar{\eta}_{\mathcal{C}} \colon \bar{\mathbb{1}} \rightarrowtail \bar{E}_{\mathcal{C}}$ we see that $\bar{E}_{\mathcal{C}} \otimes \bar{M} = 0$ forces $\bar{M} = 0$. For the second inclusion, we have by Equation (1.9) (e)

$$[-, \hat{E}_{\mathcal{C}}]_{\mathcal{A}} \cong [-, R_{\mathcal{C}}(\bar{E}_{\mathcal{C}})]_{\mathcal{A}} \cong R_{\mathcal{C}}\big([Q_{\mathcal{C}}(-), \bar{E}_{\mathcal{C}}]_{\bar{\mathcal{A}}_{\mathcal{C}}}\big).$$

It follows that $\mathcal{C}$, which is $\mathrm{Ker}(Q_{\mathcal{C}})$, is contained in $\mathrm{Ker}([-, \hat{E}_{\mathcal{C}}]_{\mathcal{A}})$ as claimed. □

We have made all the preparation for the following result.

1.24. **Theorem.** *Under Equation* (0.1)*, let $\mathcal{B} \in \mathrm{Spc}^{\mathrm{h}}(\mathcal{T}^{\mathrm{c}})$ be a maximal Serre $\otimes$-ideal of $\mathcal{A}^{\mathrm{fp}}$. Let $E_{\mathcal{B}}$ be its associated weak ring in $\mathcal{T}$ (Equation* (1.11)*) and*

$$\mathcal{B}' = \mathrm{Ker}([-, \hat{E}_{\mathcal{B}}]) = \big\{ M \in \text{Mod-}\mathcal{T}^{\mathrm{c}} \,\big|\, [M, \hat{E}_{\mathcal{B}}] = 0 \big\}, \tag{1.25}$$

where $[-,-]$ denotes the internal-hom functor in $\mathcal{A} = \text{Mod-}\mathcal{T}^{\mathrm{c}}$ (see Lemma 1.15). Then $\mathcal{B}' \subsetneq \mathcal{A}$ is a maximal localizing $\otimes$-ideal such that $\mathcal{B}' \cap \mathcal{A}^{\mathrm{fp}} = \mathcal{B}$. Furthermore, $\mathcal{B}'$ is the unique *maximal localizing $\otimes$-ideal containing $\mathcal{B}$.*

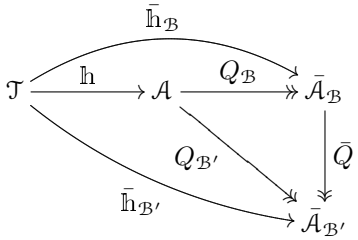
Proof. We have seen in Lemma 1.15 that such $\mathcal{B}'$ is indeed a localizing $\otimes$-ideal. As $\hat{\eta}_{\mathcal{B}} \colon \hat{\mathbb{1}} \to \hat{E}_{\mathcal{B}}$ is non-zero, $\mathcal{B}'$ is certainly proper. It contains $\mathcal{B}$ by Equation (1.23).

To show both that $\mathcal{B}'$ is maximal and unique, it suffices to show that if $\mathcal{C} \subsetneq \mathcal{A}$ is a localizing $\otimes$-ideal that contains $\mathcal{B}$ then $\mathcal{C} \subseteq \mathcal{B}'$. To see this, note that in that case $E_{\mathcal{B}} \simeq E_{\mathcal{C}}$ by Equation (1.20). Using Equation (1.23) again, we conclude that $\mathcal{C} \subseteq \mathrm{Ker}([-, \hat{E}_{\mathcal{C}}]) = \mathrm{Ker}([-, \hat{E}_{\mathcal{B}}]) = \mathcal{B}'$ as claimed. □

We can easily compare the respective residue functors of $\mathcal{B}$ and $\mathcal{B}'$.

1.26. **Proposition.** *Let $\mathcal{B} \in \mathrm{Spc}^{\mathrm{h}}(\mathcal{T}^{\mathrm{c}})$ be a homological prime and $\mathcal{B}' = \mathrm{Ker}([-, \hat{E}_{\mathcal{B}}])$ the maximal localizing $\otimes$-ideal of Equation* (1.24)*. We have a commutative diagram*

(1.27)



where $\bar{Q} \colon \bar{\mathcal{A}}_{\mathcal{B}} \twoheadrightarrow \bar{\mathcal{A}}_{\mathcal{B}'}$ is the quotient of $\bar{\mathcal{A}}_{\mathcal{B}}$ by its unique maximal localizing $\otimes$-ideal. In particular, the only localizing $\otimes$-ideals of $\bar{\mathcal{A}}_{\mathcal{B}'}$ are zero and $\bar{\mathcal{A}}_{\mathcal{B}'}$.

Proof. By Equation (1.24), the maximal localizing $\otimes$-ideal $\mathcal{B}'$ is the unique one containing $\mathcal{B}$. The Gabriel quotient $\bar{\mathcal{A}}_{\mathcal{B}'}$ of $\mathcal{A}$ by $\mathcal{B}'$ can then be done in two steps, first by $\mathrm{Loc}(\mathcal{B})$, giving us $\bar{\mathcal{A}}_{\mathcal{B}}$, and then by the maximal localizing $\otimes$-ideal of $\bar{\mathcal{A}}_{\mathcal{B}}$. □

1.28. **Proposition.** *Under the construction* $\mathcal{B} \mapsto \mathcal{B}'$ *of Equation* (1.24)*, the above support* $\operatorname{Supp}(X)$ *of an object* $X \in \mathcal{T}$ *is the following*

$$\operatorname{Supp}(X) := \big\{\mathcal{B} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{T}^c) \,\big|\, \bar{\mathbb{h}}_{\mathcal{B}'}(X) \neq 0 \text{ in } \bar{\mathcal{A}}_{\mathcal{B}'}\big\}.$$

Proof. Let $\mathcal{B} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{T}^c)$ and $\mathcal{B}' = \operatorname{Ker}([-, \hat{E}_{\mathcal{B}}])$ the corresponding maximal localizing $\otimes$-ideal of $\mathcal{A}$. By (1.5) and conservativity of restricted-Yoneda $\mathbb{h}\colon \mathcal{T} \to \mathcal{A}$, the vanishing of $[X, E_{\mathcal{B}}]_{\mathcal{T}}$ is equivalent to the vanishing of $\mathbb{h}([X, E_{\mathcal{B}}]_{\mathcal{T}}) \cong [\hat{X}, \hat{E}_{\mathcal{B}}]$ and the latter is equivalent to $\hat{X} \in \mathcal{B}'$ by (1.25), which in turn is equivalent to the vanishing of $Q_{\mathcal{B}'}(\hat{X}) = \bar{\mathbb{h}}_{\mathcal{B}'}(X)$. □

1.29. **Theorem.** *Under Equation* (0.1)*, let* A *be a weak ring in* $\mathcal{T}$ *(Equation* (0.2)*). Then we have*

$$\operatorname{Supp}(A) = \big\{\mathcal{B} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{T}^c) \,\big|\, \bar{\mathbb{h}}_{\mathcal{B}}(A) \neq 0\big\}.$$

Furthermore, if $\operatorname{Supp}(A) = \varnothing$ *then* $A = 0$.

Proof. We have done some preparation. Let $\mathcal{B} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{T}^c)$ and let $\mathcal{B}' = \operatorname{Ker}([-, E_{\mathcal{B}}])$ the corresponding maximal localizing $\otimes$-ideal. In view of Equation (1.26), we only need to show that if $\bar{\mathbb{h}}_{\mathcal{B}}(A) \neq 0$ then $\bar{\mathbb{h}}_{\mathcal{B}'}(A) \neq 0$. In Equation (1.18), we saw that if $\bar{\mathbb{h}}_{\mathcal{B}}(A) \neq 0$ then its unit $\bar{\mathbb{1}} \rightarrowtail \bar{\mathbb{h}}_{\mathcal{B}}(A)$ is a monomorphism. This uses that $\bar{\mathbb{h}}_{\mathcal{B}}(A)$ is $\otimes$-flat in $\bar{\mathcal{A}}_{\mathcal{B}}$ by Equation (1.9) (b). Applying the exact functor $\bar{Q}\colon \bar{\mathcal{A}}_{\mathcal{B}} \to \bar{\mathcal{A}}_{\mathcal{B}'}$ coming from Equation (1.26), we have a monomorphism $\bar{\mathbb{1}} \rightarrowtail \bar{\mathbb{h}}_{\mathcal{B}'}(A)$ in $\bar{\mathcal{A}}_{\mathcal{B}'}$, forcing $\bar{\mathbb{h}}_{\mathcal{B}'}(A) \neq 0$. For the 'furthermore' part, suppose that $\operatorname{Supp}(A) = \varnothing$, i.e., that $\bar{\mathbb{h}}_{\mathcal{B}}(A) = 0$ for all $\mathcal{B} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{T}^c)$. Then we can invoke the Nilpotence Theorem of [4] for the unit $\eta\colon \mathbb{1} \to A$ of the weak ring A. Clearly, $\bar{\mathbb{h}}_{\mathcal{B}}(\eta) = 0$ for all $\mathcal{B}$, since the target of these maps are all zero. Hence $\eta^{\otimes n} = 0$ for some $n \gg 1$. However, $A \otimes \eta\colon A \rightarrowtail A \otimes A$ is a split monomorphism by definition of a weak ring, hence $A \otimes \eta^{\otimes n}\colon A \rightarrowtail A^{\otimes n+1}$ is a composite of split monomorphisms. Thus $\eta^{\otimes n} = 0$ forces $A = 0$ as claimed. □

1.30. **Recollection.** We consider a tensor-triangulated functor $F\colon \mathcal{T} \to \mathcal{S}$ between big tt-categories as in Equation (0.1). Because every tensor functor preserves rigidity and because we assume that rigid and compact objects coincide, F restricts to a tt-functor $F\colon \mathcal{T}^c \to \mathcal{S}^c$. We assume that our functor F preserves coproducts. By Brown-Neeman Representability [21], this is equivalent to the existence of a right adjoint $U\colon \mathcal{S} \to \mathcal{T}$, that is then lax-monoidal since F is monoidal. This implies that $U(\mathbb{1})$ is a commutative ring object in $\mathcal{T}$. In fact, since F preserves compacts, U preserves coproducts (and thus admits another right adjoint).

By Krause [19, Cor. 2.4], both F and U induce exact coproduct-preserving functors $\hat{F}\colon \text{Mod-}\mathcal{T}^c \to \text{Mod-}\mathcal{S}^c$ and $\hat{U}\colon \text{Mod-}\mathcal{S}^c \to \text{Mod-}\mathcal{T}^c$ such that $\hat{F} \circ \mathbb{h}_{\mathcal{T}} \cong \mathbb{h}_{\mathcal{S}} \circ F$ and $\hat{U} \circ \mathbb{h}_{\mathcal{S}} \cong \mathbb{h}_{\mathcal{T}} \circ \hat{U}$. These can be described by the usual Kan formulas

$$\hat{F}(M) = \operatorname*{colim}_{(\hat{x} \to M) \in (\mathbb{h}_{\mathcal{T}}/M)} \mathbb{h}_{\mathcal{S}}(F(x)) \qquad \text{and} \qquad \hat{U}(N) = \operatorname*{colim}_{(\hat{y} \to N) \in (\mathbb{h}_{\mathcal{S}}/N)} \mathbb{h}_{\mathcal{T}}(U(y)).$$

Alternatively, when viewing $N \in \text{Mod-}\mathcal{S}^c$ as an additive functor $(\mathcal{S}^c)^{\mathrm{op}} \to \mathrm{Ab}$, we have $\hat{U}(N) = N \circ F\colon (\mathcal{T}^c)^{\mathrm{op}} \to \mathrm{Ab}$. Since $\otimes$ is colimit-preserving and F is symmetric monoidal, it follows that $\hat{F}$ is also symmetric monoidal. The adjunction $F \dashv U$ yields an adjunction $\hat{F} \dashv \hat{U}$ making the following diagram of adjunctions commute:

$$\begin{array}{ccc} \mathcal{T} & \xrightarrow{\mathbb{h}_{\mathcal{T}}} & \text{Mod-}\mathcal{T}^c \\ F \big\downarrow \big\uparrow U & & \hat{F} \big\downarrow \big\uparrow \hat{U} \\ \mathcal{S} & \xrightarrow{\mathbb{h}_{\mathcal{S}}} & \text{Mod-}\mathcal{S}^c. \end{array} \tag{1.31}$$

Note also that $\hat{F}$ preserves finitely presented objects $\hat{F}(\text{mod-}\mathcal{T}^c) \subseteq \text{mod-}\mathcal{S}^c$ since they are generated by the compact objects. See details in [14, Constr. 6.10]. Finally, since F and U satisfy a projection formula $U(Y \otimes F(X)) \cong U(Y) \otimes X$ by [6, 2.16], the same holds for $\hat{F}$ and $\hat{U}$:

$$\hat{U}(N \otimes \hat{F}(M)) \cong \hat{U}(N) \otimes M \tag{1.32}$$

by using that $\hat{F}$, $\hat{U}$ and $\otimes$ commute with colimits and the above explicit formulas.

1.33. **Remark.** We want to show functoriality of $\mathrm{Spc}^{\mathrm{h}}(-)$. The following result is not entirely obvious, as the commutative algebraists will recognize. For a homomorphism of commutative rings, it is not true in general that the preimage of a maximal ideal is maximal. This vindicates again the use of *maximal* ideals $\mathcal{B}$ in mod-$\mathcal{T}^c$ when constructing $\mathrm{Spc}^{\mathrm{h}}(\mathcal{T}^c)$, as opposed to some kind of prime ideals.

The following is extracted from [4]:

1.34. **Proposition.** *Let $\mathcal{T}$ be a big tt-category as in Equation* (0.1)*. Let $H\colon \mathcal{T} \to \mathcal{D}$ be a homological functor to a locally coherent Grothendieck category $\mathcal{D}$ admitting a colimit-preserving tensor. Assume that H is monoidal*[1] *and maps compact objects of $\mathcal{T}$ to finitely presented objects. Assume furthermore that $H(X)$ is $\otimes$-flat in $\mathcal{D}$ for every $X \in \mathcal{T}$. Let $\hat{H}\colon \mathcal{A} = \text{Mod-}\mathcal{T}^c \to \mathcal{D}$ be the exact coproduct-preserving functor induced by H. Then $\mathrm{Ker}(\hat{H})$ is a localizing $\otimes$-ideal generated by its finitely presented part $\mathcal{B} := \mathrm{Ker}(\hat{H}) \cap \mathcal{A}^{\mathrm{fp}}$. If moreover $\mathcal{D}^{\mathrm{fp}}$ is simple, i.e., has only zero and $\mathcal{D}^{\mathrm{fp}}$ as Serre $\otimes$-ideals, then $\mathcal{B}$ is a homological prime, i.e., it is maximal in $\mathcal{A}^{\mathrm{fp}}$.*

Proof. This is the first page of the proof of [4, Th. 5.6]. As the notation in *loc. cit.* depends on some triangular primes $\mathcal{P}$, we outline a cleaned-up version here for the reader's convenience. The kernel $\mathrm{Ker}(\hat{H}\colon \mathcal{A} \to \mathcal{D}) = \mathrm{Loc}(\mathcal{B})$ is generated by its finitely presented part $\mathcal{B}$ by [14, Prop. A.6]. To show that $\mathcal{B}$ is maximal when $\mathcal{D}^{\mathrm{fp}}$ is simple, assume that $\mathcal{C} \supsetneq \mathcal{B}$ is larger and show $\mathcal{C} = \mathcal{A}$. For that, pick $M \in \mathcal{C} \smallsetminus \mathcal{B}$, so that $M \otimes \hat{E}_{\mathcal{B}} \neq 0$ whereas $M \otimes \hat{E}_{\mathcal{C}} = 0$. From the former deduce that $\hat{H}(M) \neq 0$ in $\mathcal{D}^{\mathrm{fp}}$ and from the latter deduce that $\hat{H}(M) \otimes H(E_{\mathcal{C}}) = 0$ in $\mathcal{D}$. Use simplicity of $\mathcal{D}^{\mathrm{fp}}$ and $\otimes$-flatness of $H(E_{\mathcal{C}})$ to deduce that $H(E_{\mathcal{C}}) = 0$, i.e., $\hat{E}_{\mathcal{C}} \in \mathrm{Ker}(\hat{H}) = \mathrm{Loc}(\mathcal{B}) \subset \mathrm{Loc}(\mathcal{C})$, forcing $\hat{\mathbb{1}} \in \mathcal{C}$ as well, from the exact sequence $J \rightarrowtail \hat{\mathbb{1}} \to \hat{E}_{\mathcal{C}}$ where $J \in \mathrm{Loc}(\mathcal{C})$, as always, and $E_{\mathcal{C}} \in \mathrm{Loc}(\mathcal{C})$ as just proved. □

1.35. **Lemma.** *With notation as in Equation* (1.30)*, let $\mathcal{C} \in \mathrm{Spc}^{\mathrm{h}}(\mathcal{S}^c)$. Then $\mathcal{B} := \big\{M \in \text{mod-}\mathcal{T}^c \,\big|\, \hat{F}(M) \in \mathcal{C}\big\}$ is a maximal Serre $\otimes$-ideal of* mod-$\mathcal{T}^c$*. Furthermore, we have a diagram*

$$\begin{array}{ccccc}
\mathcal{T} & \xrightarrow{\mathbb{h}_{\mathcal{T}}} & \mathcal{A} = \text{Mod-}\mathcal{T}^c & \underset{Q_{\mathcal{B}}}{\overset{R_{\mathcal{B}}}{\rightleftarrows}} & \bar{\mathcal{A}}_{\mathcal{B}} = \text{Mod-}\mathcal{T}^c/\,\mathrm{Loc}(\mathcal{B}) \\
F \big\downarrow \big\uparrow U & & \hat{F} \big\downarrow \big\uparrow \hat{U} & & \bar{F} \big\downarrow \big\uparrow \bar{U} \\
\mathcal{S} & \xrightarrow{\mathbb{h}_{\mathcal{S}}} & \text{Mod-}\mathcal{S}^c & \underset{Q_{\mathcal{C}}}{\overset{R_{\mathcal{C}}}{\rightleftarrows}} & \text{Mod-}\mathcal{S}^c/\,\mathrm{Loc}(\mathcal{C})
\end{array} \tag{1.36}$$

with $\mathbb{h}_{\mathcal{S}}\, F \cong \hat{F}\, \mathbb{h}_{\mathcal{T}}$ and $\mathbb{h}_{\mathcal{T}}\, U \cong \hat{U}\, \mathbb{h}_{\mathcal{S}}$ and $Q_{\mathcal{C}}\, \hat{F} \cong \bar{F}\, Q_{\mathcal{B}}$ and $\hat{U}\, R_{\mathcal{C}} \cong R_{\mathcal{B}}\, \bar{U}$. Finally, the pure-injective $E_{\mathcal{B}}$ is a direct summand of $U(E_{\mathcal{C}})$ in $\mathcal{T}$ (see Equation (1.11)*).*

[1] not necessarily *symmetric* monoidal.

Proof. The left-hand square of (1.36) is (1.31), repeated for cognitive help. Since $\hat{F}$ is exact, $\hat{U}$ preserves injectives hence $U(E_{\mathcal{C}})$ is pure-injective. Since U is lax-monoidal, $U(E_{\mathcal{C}})$ is a weak ring, with unit $\mathbb{1} \to U(E_{\mathcal{C}})$ adjoint to $\eta_{\mathcal{C}} \colon \mathbb{1}_{\mathcal{S}} = F(\mathbb{1}_{\mathcal{T}}) \to E_{\mathcal{C}}$. In particular, $U(E_{\mathcal{C}}) \neq 0$. The internal-hom version of the adjunction reads

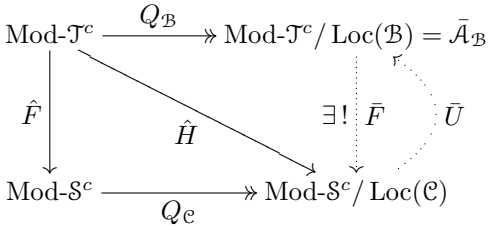
$$[M, \hat{U}(\hat{E}_{\mathcal{C}})] \cong \hat{U}[\hat{F}(M), \hat{E}_{\mathcal{C}}] \tag{1.37}$$

for every $M \in \mathcal{A}$, as can be checked by testing under $\mathrm{Hom}_{\mathcal{A}}(\hat{x}, -)$ for $x \in \mathcal{T}^c$.

Consider now the functor $H := Q_{\mathcal{C}} \circ \hat{F} \circ \mathbb{h}_{\mathcal{T}}$

$$H \colon \mathcal{T} \to \text{Mod-}\mathcal{T}^c \to \text{Mod-}\mathcal{S}^c \twoheadrightarrow \text{Mod-}\mathcal{S}^c/\operatorname{Loc}(\mathcal{C}) =: \mathcal{D}.$$

This is a homological functor to which we can apply Equation (1.34), with $\hat{H}$ being necessarily $Q_{\mathcal{C}} \circ \hat{F}$. It tells us that $\mathcal{B} = \operatorname{Ker}(\hat{H}) \cap \mathcal{A}^{\mathrm{fp}}$ is indeed a maximal Serre $\otimes$-ideal of $\mathcal{A}^{\mathrm{fp}}$. We can then factor $\hat{H}$ via $Q_{\mathcal{B}}$, yielding a unique tensor-exact coproduct-preserving functor $\bar{F} \colon \text{Mod-}\mathcal{T}^c/\operatorname{Loc}(\mathcal{B}) \to \text{Mod-}\mathcal{S}^c/\operatorname{Loc}(\mathcal{C})$ making the following diagram commute



By general Grothendieck-category theory, $\bar{F}$ admits a right adjoint $\bar{U}$. This gives us (1.36). Note that $R_{\mathcal{B}}\bar{U}$ is the right adjoint to $\bar{F}\, Q_{\mathcal{B}} \cong Q_{\mathcal{C}}\, \hat{F}$, whose right adjoint is also $\hat{U}\, R_{\mathcal{C}}$. Hence we have

$$R_{\mathcal{B}} \circ \bar{U} \cong \hat{U} \circ R_{\mathcal{C}}. \tag{1.38}$$

Consider $\bar{\mathbb{1}} \rightarrowtail \bar{E}_{\mathcal{C}}$ the injective hull of the unit in Mod-$\mathcal{S}^c/\operatorname{Loc}(\mathcal{C})$, as in the statement. The pure-injective $E_{\mathcal{C}} \in \mathcal{S}$ corresponding to $\mathcal{C}$ is then characterized by $\hat{E}_{\mathcal{C}} = R_{\mathcal{C}}(\bar{E}_{\mathcal{C}})$. Therefore, by (1.38), we have $R_{\mathcal{B}}(\bar{U}(\bar{E}_{\mathcal{C}})) \cong \hat{U}(\hat{E}_{\mathcal{C}}) = \mathbb{h}_{\mathcal{T}}(U(E_{\mathcal{C}}))$. Applying $Q_{\mathcal{B}}$ to this relation, we see that the injective $\bar{U}(\bar{E}_{\mathcal{C}})$ is the image in $\bar{\mathcal{A}}_{\mathcal{B}}$ of $U(E_{\mathcal{C}})$. By Equation (1.18) the unit of this non-trivial weak ring $\bar{U}(\bar{E}_{\mathcal{C}})$ is a monomorphism $\bar{\mathbb{1}} \rightarrowtail \bar{U}(\bar{E}_{\mathcal{C}})$ in $\bar{\mathcal{A}}_{\mathcal{B}}$ and $\bar{U}(\bar{E}_{\mathcal{C}})$ is injective. Therefore the injective hull $\bar{E}_{\mathcal{B}}$ of $\bar{\mathbb{1}}$ is a direct summand of $\bar{U}(\bar{E}_{\mathcal{C}})$. As usual, this decomposition holds in $\mathcal{T}$ already, by Equation (1.9) (d). □

Summarizing our discussion:

1.39. **Theorem.** *Let $F \colon \mathcal{T} \to \mathcal{S}$ be a coproduct-preserving tt-functor between big tt-categories. Then the map* $\operatorname{Spc}^{\mathrm{h}}(F) \colon \operatorname{Spc}^{\mathrm{h}}(\mathcal{S}^c) \to \operatorname{Spc}^{\mathrm{h}}(\mathcal{T}^c)$

$$\begin{aligned} \operatorname{Spc}^{\mathrm{h}}(\mathcal{S}^c) &\to \operatorname{Spc}^{\mathrm{h}}(\mathcal{T}^c) \\ \mathcal{C} &\longmapsto \hat{F}^{-1}(\mathcal{C}) \end{aligned}$$

is well-defined and makes the following square commute

$$\begin{array}{ccc} \operatorname{Spc}^{\mathrm{h}}(\mathcal{S}^c) & \xrightarrow{\operatorname{Spc}^{\mathrm{h}}(F)} & \operatorname{Spc}^{\mathrm{h}}(\mathcal{T}^c) \\ {\scriptstyle \phi_{\mathcal{S}}}\downarrow & & \downarrow{\scriptstyle \phi_{\mathcal{T}}} \\ \operatorname{Spc}(\mathcal{S}^c) & \xrightarrow{\operatorname{Spc}(F)} & \operatorname{Spc}(\mathcal{T}^c) \end{array} \tag{1.40}$$

where $\phi \colon \operatorname{Spc}^{\mathrm{h}} \to \operatorname{Spc}$ is as in Equation (0.3).

Proof. We use Equation (1.35) to show that the map $\mathrm{Spc}^{\mathrm{h}}(F)$ is well-defined. The commutativity of the square then follows from $\hat{F} \circ \mathbb{h}_{\mathcal{T}} \cong \mathbb{h}_{\mathcal{S}} \circ F \colon \mathcal{T}^c \to \text{Mod-}\mathcal{S}^c$. □

1.41. **Theorem.** *Let $F \colon \mathcal{T} \to \mathcal{S}$ be a tt-functor between big tt-categories as in Equation* (0.1) *with a right adjoint $U \colon \mathcal{S} \to \mathcal{T}$ (Equation* (1.30)*). Then the image of* $\mathrm{Spc}^{\mathrm{h}}(F) \colon \mathrm{Spc}^{\mathrm{h}}(\mathcal{S}^c) \to \mathrm{Spc}^{\mathrm{h}}(\mathcal{T}^c)$ *is equal to* $\mathrm{Supp}(U(\mathbb{1}))$.

Proof. Let $\psi = \mathrm{Spc}^{\mathrm{h}}(F) \colon \mathrm{Spc}^{\mathrm{h}}(\mathcal{S}^c) \to \mathrm{Spc}^{\mathrm{h}}(\mathcal{T}^c)$ and $U(\mathbb{1}) \in \mathcal{T}$. Since $U(\mathbb{1})$ is a ring object, we have $\mathrm{Supp}(U(\mathbb{1})) = \big\{\mathcal{B} \in \mathrm{Spc}^{\mathrm{h}}(\mathcal{T}^c) \,\big|\, \bar{\mathbb{h}}_{\mathcal{B}}(U(\mathbb{1})) \neq 0\big\}$ by Equation (1.29).

The inclusion $\mathrm{Im}(\psi) \subseteq \mathrm{Supp}(U(\mathbb{1}))$ is relatively easy. Suppose that $\mathcal{B}$ belongs to the image of ψ and let $\mathcal{C} \in \mathrm{Spc}^{\mathrm{h}}(\mathcal{S}^c)$ such that $\mathcal{B} = \hat{F}^{-1}(\mathcal{C})$. We apply Equation (1.35) and we have in particular the diagram (1.36). The lax-monoidal functor U maps $\mathbb{1}_{\mathcal{S}}$ to a commutative ring object $U(\mathbb{1})$ which acts on every object of the form $U(Y)$, since $\mathbb{1}_{\mathcal{S}}$ acts on every object $Y \in \mathcal{S}$. (This action is as a 'module' but we avoid this terminology since we already mean something else by 'modules'.) We apply this to $Y = E_{\mathcal{C}}$ the pure-injective object corresponding to $\mathcal{C}$. So, the ring $U(\mathbb{1})$ acts on $U(E_{\mathcal{C}})$ in $\mathcal{T}$. Applying the tensor functor $\bar{\mathbb{h}}_{\mathcal{B}} \colon \mathcal{T} \to \bar{\mathcal{A}}_{\mathcal{B}}$, we see that the ring object $\bar{\mathbb{h}}_{\mathcal{B}}(U(\mathbb{1}))$ acts on $\bar{\mathbb{h}}_{\mathcal{B}}(U(E_{\mathcal{C}}))$ in $\bar{\mathcal{A}}_{\mathcal{B}}$. In particular if, *ab absurdo*, the ring $\bar{\mathbb{h}}_{\mathcal{B}}(U(\mathbb{1}))$ vanished then so would the object $\bar{\mathbb{h}}_{\mathcal{B}}(U(E_{\mathcal{C}}))$. By Equation (1.35), we also know that $E_{\mathcal{B}}$ is a direct summand of $U(E_{\mathcal{C}})$, hence $\bar{\mathbb{h}}_{\mathcal{B}}(E_{\mathcal{B}})$ would be a direct summand of $\bar{\mathbb{h}}_{\mathcal{B}}(U(E_{\mathcal{C}})) = 0$. This implies the vanishing of the injective hull $\bar{\mathbb{h}}_{\mathcal{B}}(E_{\mathcal{B}}) = \bar{E}_{\mathcal{B}}$ of $\bar{\mathbb{1}}$ in $\bar{\mathcal{A}}_{\mathcal{B}}$, a contradiction. So $\bar{\mathbb{h}}_{\mathcal{B}}(U(\mathbb{1}))$ cannot be zero, meaning $\mathcal{B} \in \mathrm{Supp}(U(\mathbb{1}))$.

Conversely, let $\mathcal{B}$ be such that $\bar{\mathbb{h}}_{\mathcal{B}}(U(\mathbb{1})) \neq 0$ and let us show that $\mathcal{B} \in \mathrm{Im}(\psi)$. Let I be the kernel of $\hat{\eta}_{\mathcal{B}} \colon \hat{\mathbb{1}} \to \hat{E}_{\mathcal{B}}$ in $\mathcal{A}$. We have an exact sequence $I \rightarrowtail \hat{\mathbb{1}} \to \hat{E}_{\mathcal{B}}$ and therefore $\hat{F}(I) \rightarrowtail \hat{F}(\hat{\mathbb{1}}) \to \hat{F}(\hat{E}_{\mathcal{B}})$ in Mod-$\mathcal{S}^c$. Note that $\hat{F}(\hat{E}_{\mathcal{B}}) = \widehat{F(E_{\mathcal{B}})}$ is flat. Consider in mod-$\mathcal{S}^c$ the Serre $\otimes$-ideal

$$\mathcal{C}_0 := \mathrm{Ker}\left((\hat{F}(\hat{E}_{\mathcal{B}}) \otimes -)_{|_{\text{mod-}\mathcal{S}^c}}\right) = \big\{N \in \text{mod-}\mathcal{S}^c \,\big|\, \hat{F}(\hat{E}_{\mathcal{B}}) \otimes N = 0\big\}.$$

If, *ab absurdo*, $\mathcal{C}_0 = \text{mod-}\mathcal{S}^c$ then $\hat{F}(\hat{E}_{\mathcal{B}}) = 0$, hence $\hat{F}(I) = \hat{\mathbb{1}}$. Since we have $I = \mathrm{colim}_{\text{f.p. } M \rightarrowtail I} M$ by Equation (1.14) (c), and since $\hat{\mathbb{1}}_{\mathcal{S}}$ is finitely presented, there exists $M \rightarrowtail I$ finitely presented such that $\hat{F}(M) = \hat{\mathbb{1}}$ already. Note that $M \rightarrowtail I \in \mathrm{Loc}(\mathcal{B})$ implies $M \in \mathcal{B}$. Now, we use $\hat{U}\hat{F} \cong \hat{U}(\mathbb{1}) \otimes -$ by the projection formula (1.32). In short we have $M \otimes \hat{U}(\mathbb{1}) \cong \hat{U}(\mathbb{1})$ with $M \in \mathcal{B} \subset \mathrm{Ker}(\hat{E}_{\mathcal{B}} \otimes -)$. This implies that $\hat{E}_{\mathcal{B}} \otimes \hat{U}(\mathbb{1}) \cong \hat{E}_{\mathcal{B}} \otimes M \otimes \hat{U}(\mathbb{1}) \cong 0$ and thus $E_{\mathcal{B}} \otimes U(\mathbb{1}) = 0$ meaning that $\mathcal{B} \notin \mathrm{Supp}(U(\mathbb{1}))$, a contradiction. So $\mathcal{C}_0$ is a proper $\otimes$-ideal of mod-$\mathcal{S}^c$. Then choose any maximal $\otimes$-ideal $\mathcal{C} \in \mathrm{Spc}^{\mathrm{h}}(\mathcal{S}^c)$ containing $\mathcal{C}_0$. Then $\hat{F}^{-1}(\mathcal{C}) \supseteq \mathcal{B}$ by construction hence by maximality $\mathcal{B} = \hat{F}^{-1}(\mathcal{C}) = \psi(\mathcal{C})$ and indeed $\mathcal{B} \in \mathrm{Im}(\psi)$. □

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